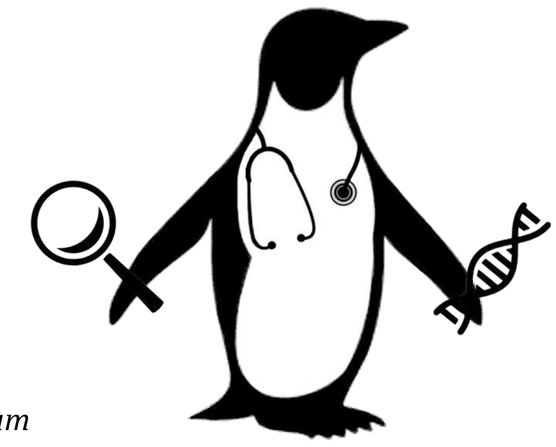
PENGQUIN:
PErsoNal Genome QUery IN Healthcare and Clinical Practice



Elias Crum[1,2]

[1]*IDLab, Department of Electronics and Information Systems, Ghent University – imec, Belgium*
[2]*Flemish institute for Technological Research (VITO) Mol, Belgium*

Genome Data for Clinical Care?

Still Many challenges:

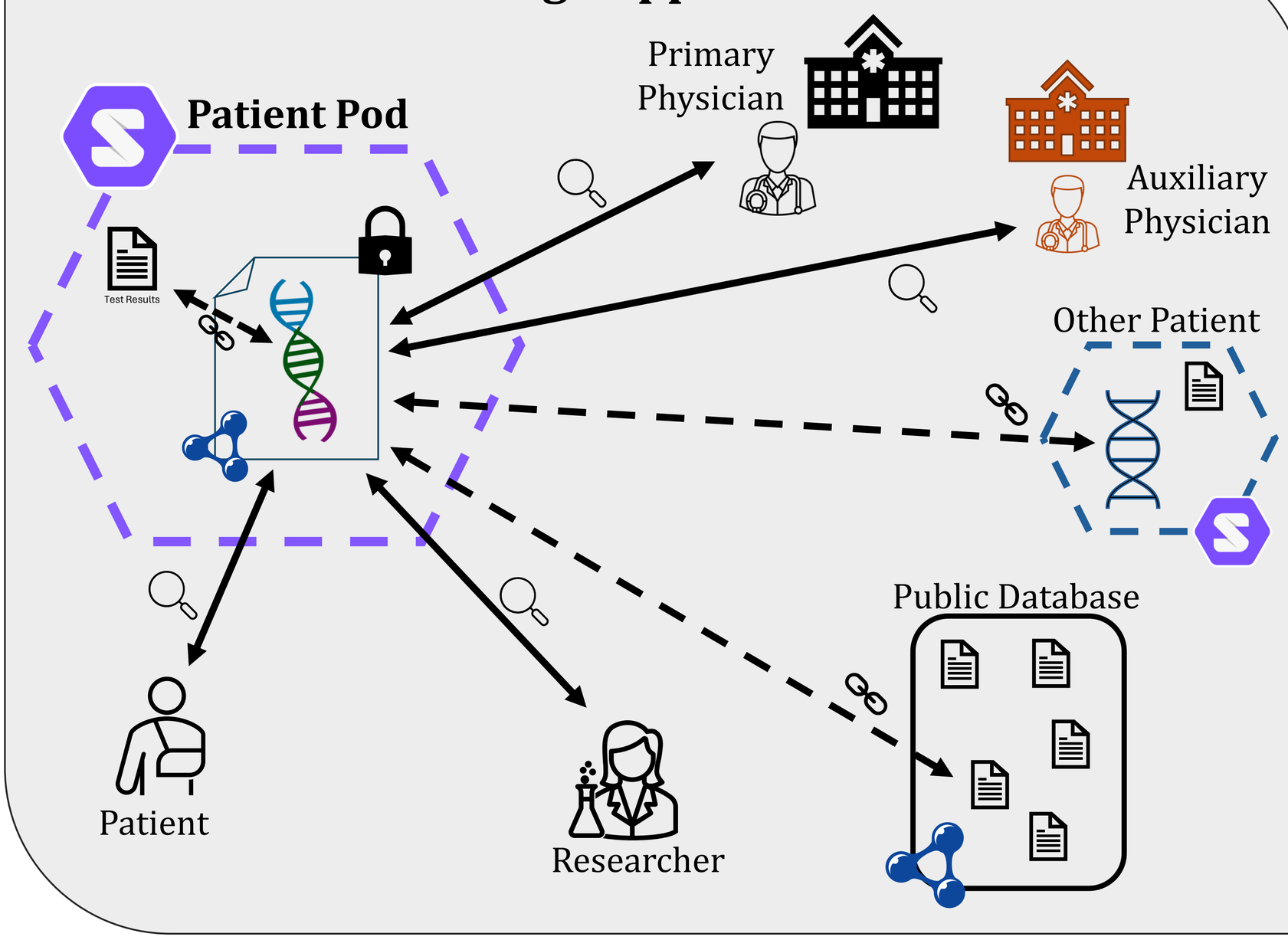
- Storage
- Privacy
- Governance
- Formatting

Could Semantic Web technology help?

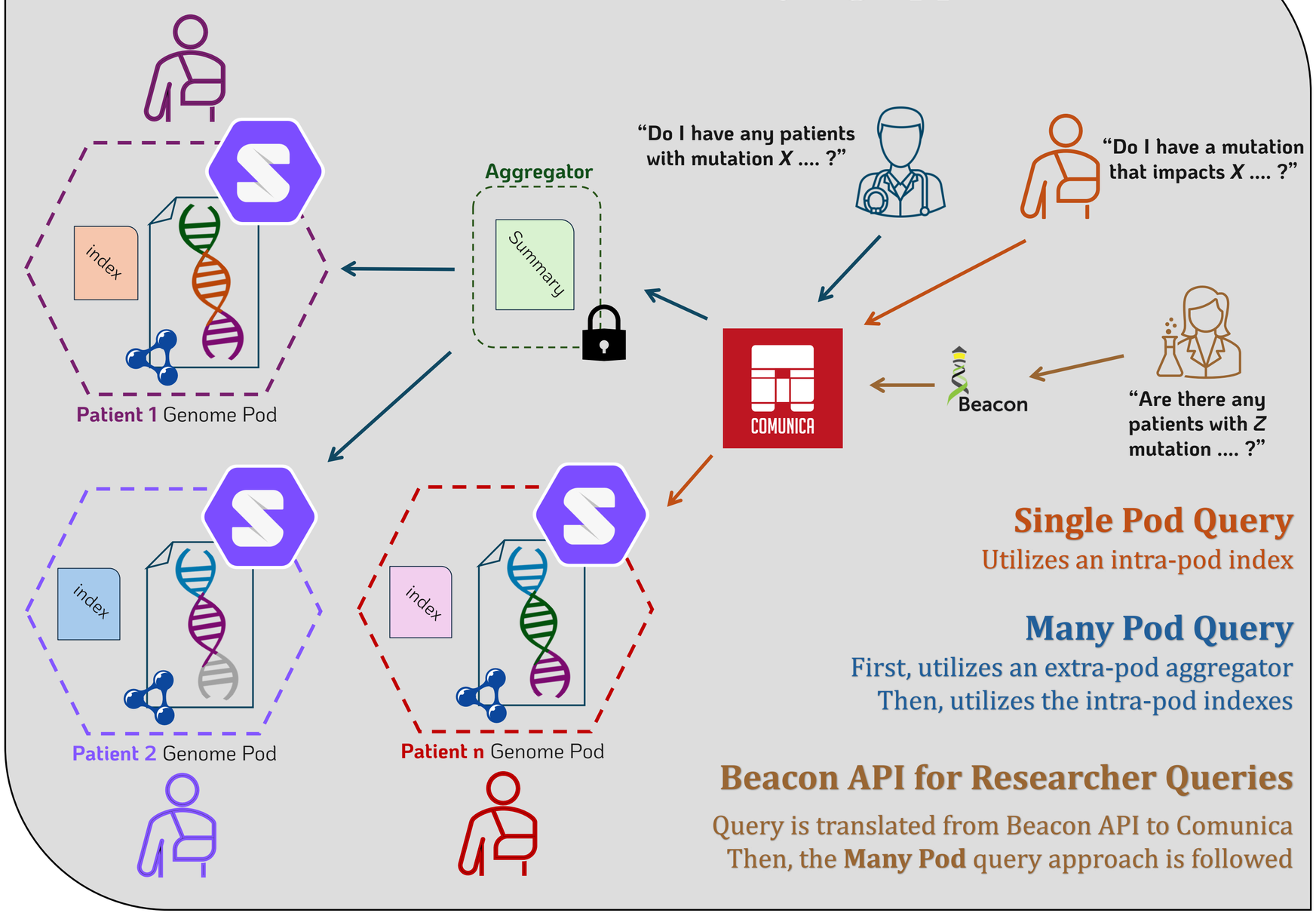
Personal Genome Sequences

- Variant Call Format (VCF)
- Very large (~GBs)
- Strict privacy protections

A Patient-Centric Storage Approach



An Index-Based Link Traversal Querying Approach



Single Pod Query
Utilizes an intra-pod index

Many Pod Query
First, utilizes an extra-pod aggregator
Then, utilizes the intra-pod indexes

Beacon API for Researcher Queries
Query is translated from Beacon API to Comunica
Then, the **Many Pod** query approach is followed

Research Challenges

- Privacy granularity
  - Index privacy concerns?
  - How granular is adequate?
- RDF Conversion + Data Linkages
  - VCF to RDF Conversion?
  - What data should be linked?
  - Automation of data linking?
- Querying performance
  - Many, large data sources
  - Algorithms that use indexes?

Engineering Tasks

- Genomic data storage in Solid pods
- Data privacy management
- Data Sharing via the web

Research Pursuits

- Represent genomic data as RDF?
- Linking genomic and health data?
- Single and Many pod Querying?

Ph.D. Symp. Paper

Contact

PhD Promoters: Ruben Taelman[1], Bart Buelens[2], Gökhan Ertaylan[2] and Ruben Verborgh[1] | Funding provided from VITO NV (**UG_PhD_2303_contract**)